

WAUPACA, WI, USA

WMO: 720343

Lat: 44.333N

Lon: 89.020W

Elev: 826

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54852

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.0	-3.2	-17.4	2.2	-6.2	-13.0	2.8	-1.7	20.8	21.0	18.6	19.3	5.1	280	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	90.4	72.3	87.4	71.0	84.4	69.1	75.6	84.6	73.9	82.5	72.3	80.5	9.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	126.5	79.6	71.5	120.1	78.4	69.8	113.1	76.5	39.6	84.7	38.0	82.7	36.5	80.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	17.0	15.2	DB	-14.5	95.1	4.3	2.9	-17.6	97.2	-20.1	98.9	-22.5	100.6	-25.6	102.7	(4)
(5)			WB	-15.1	79.1	4.1	2.1	-18.1	80.5	-20.5	81.7	-22.8	82.9	-25.8	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	19.2	21.3	33.4	45.3	57.9	67.4	71.7	69.7	61.9	49.2	36.4	24.0		
	DBStd	20.71	11.67	11.15	11.63	8.95	8.97	6.28	5.85	5.50	7.97	8.64	9.44	10.75		
	HDD50	3864	956	804	531	193	25	0	0	0	7	125	418	805		
	HDD65	7325	1421	1224	982	592	258	45	11	17	150	496	859	1270		
	CDD50	2617	0	0	15	52	270	523	673	612	364	99	9	0		
	CDD65	604	0	0	1	2	38	118	219	164	57	5	0	0		
	CDH74	6068	0	0	15	51	482	1155	2247	1485	570	63	0	0		
	CDH80	1936	0	0	1	9	147	343	827	445	158	6	0	0		
	WSAvg	5.9	6.4	6.6	6.5	7.4	6.6	5.5	4.7	4.3	4.8	6.0	6.3	6.3		
	PrecAvg	32.2	1.3	1.1	1.9	3.1	4.0	4.4	3.7	3.9	3.2	2.5	1.8	1.5		
(15) Precipitation	PrecMax	39.1	3.1	1.9	3.6	5.4	6.6	8.4	9.2	11.0	7.6	5.7	5.0	3.2		
	PrecMin	26.6	0.4	0.3	0.3	0.8	2.2	1.5	1.1	1.1	0.5	0.4	0.0	0.3		
	PrecStd	3.5	0.7	0.5	1.0	1.2	1.3	2.2	2.3	2.0	1.7	1.3	1.3	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	54.9	74.9	80.8	90.1	91.7	96.9	91.7	90.2	81.0	68.3	51.6		
		MCWB	40.0	47.1	61.4	59.5	68.2	72.0	75.2	72.5	73.1	67.7	56.1	44.1		
	2%	DB	40.8	44.6	63.1	72.5	83.9	87.7	90.6	88.0	84.9	74.7	60.1	44.1		
		MCWB	36.5	39.0	52.4	56.3	65.6	69.8	72.9	71.7	70.2	63.7	51.5	40.5		
	5%	DB	37.2	39.3	55.8	67.4	79.6	83.6	87.7	84.9	80.7	69.5	54.8	40.8		
		MCWB	34.9	35.3	47.4	53.1	63.0	68.2	71.6	69.8	67.2	60.4	48.3	38.2		
	10%	DB	34.6	36.0	49.9	62.1	74.8	80.9	84.5	81.9	76.7	64.0	50.2	37.1		
		MCWB	32.9	32.9	43.0	49.8	61.3	66.5	69.9	67.7	65.7	56.5	45.2	34.8		
	0.4%	WB	41.0	46.7	61.8	63.4	72.5	77.0	79.4	76.9	75.5	71.8	58.0	49.0		
		MCDB	45.8	54.9	74.1	73.3	83.9	86.5	90.4	85.2	84.2	78.0	64.3	50.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.6	39.6	54.8	59.1	69.2	73.6	76.1	74.8	72.8	66.7	53.1	41.3		
		MCDB	40.1	43.6	60.2	69.5	77.6	83.7	85.7	82.6	80.4	70.5	57.9	43.2		
	5%	WB	35.2	35.9	48.4	54.9	66.5	71.4	74.1	72.9	70.7	61.9	49.2	38.1		
		MCDB	37.0	39.4	56.2	64.3	75.9	79.7	83.0	80.4	77.1	67.9	54.5	40.4		
	10%	WB	33.0	32.9	43.0	50.6	63.5	69.2	72.5	71.2	68.1	57.5	45.5	35.0		
		MCDB	34.5	35.2	49.4	60.5	71.9	76.2	81.0	78.4	73.8	63.1	49.0	36.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	16.6	18.6	20.4	22.3	21.3	23.0	21.5	22.3	18.9	15.5	13.2		
		MCDBR	16.4	18.5	26.0	31.4	28.1	25.2	25.8	24.2	26.4	26.6	24.4	15.4		
		MCWBR	13.6	14.0	15.8	16.5	12.9	11.2	10.6	9.8	12.8	16.0	16.2	13.0		
		MCDBR	15.2	17.3	24.7	27.9	23.6	21.5	22.4	20.0	21.3	22.7	22.1	14.0		
	5% WB	MCWBR	13.0	13.3	16.0	16.7	12.2	11.4	11.1	9.7	11.7	16.2	16.1	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.309	0.327	0.364	0.399	0.403	0.392	0.393	0.364	0.324	0.293	0.266		
	taud		2.400	2.237	2.366	2.330	2.275	2.298	2.337	2.327	2.403	2.511	2.518	2.443		
	Ebn at Noon		279	277	289	285	277	275	277	272	272	272	264	264		
	Edh at Noon		23	33	34	38	42	41	39	38	32	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		527	871	1233	1442	1728	1892	1937	1656	1319	835	534	390		
	RadStd		60	79	133	145	134	136	94	109	101	89	53	41		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page